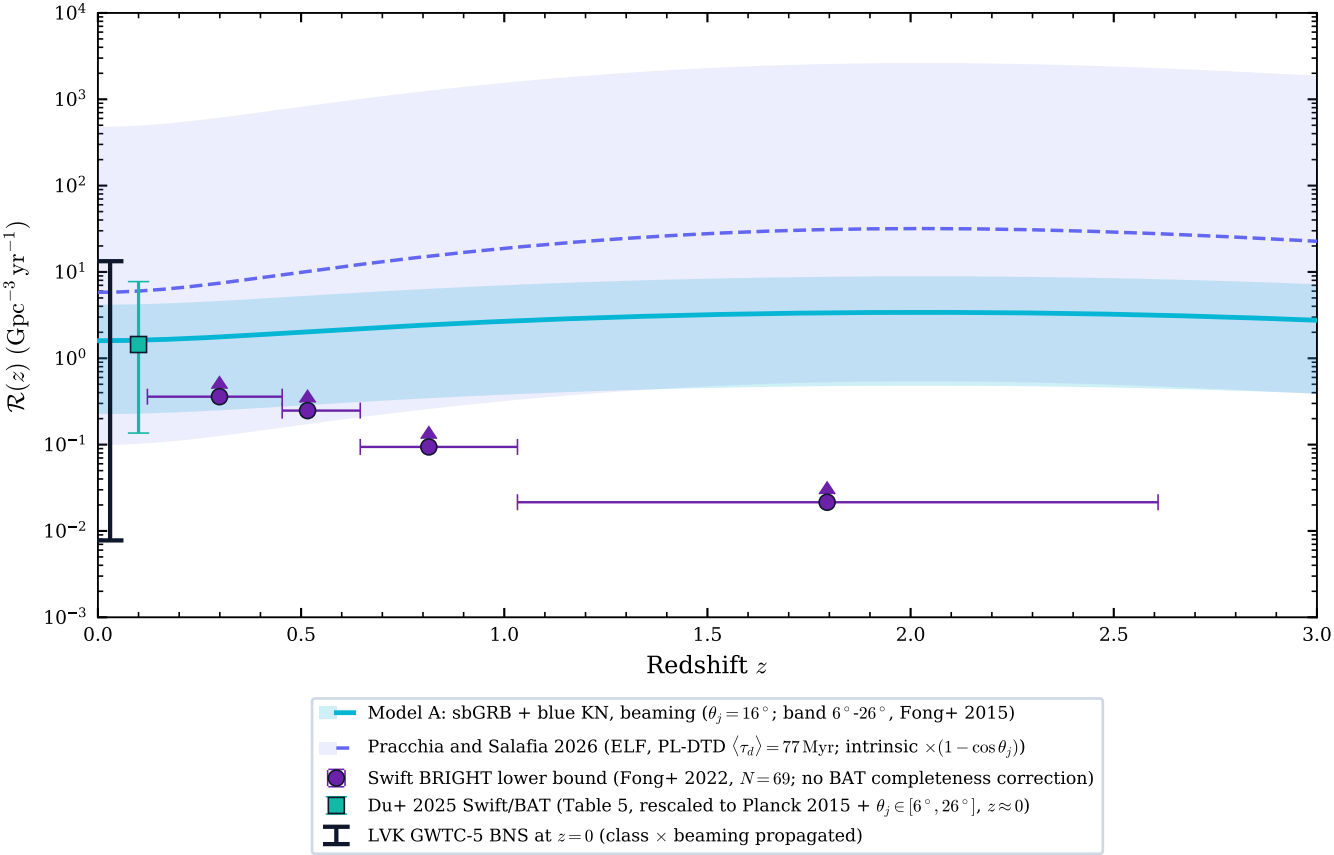

10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
$\mathcal{R}(z)$ (Gpc^{-3} yr^{-1})
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Redshift z
Model A: sbGRB + blue KN, beaming ($\theta_j = 16^\circ$; band 6°-26°, Fong+ 2015)
Pracchia and Salafia 2026 (ELF, PL-DTD $\langle \tau_d \rangle = 77$ Myr; intrinsic $\times(1-\cos\theta_j)$)
Swift BRIGHT lower bound (Fong+ 2022, $N = 69$; no BAT completeness correction)
Du+ 2025 Swift/BAT (Table 5, rescaled to Planck 2015 + $\theta_j \in [6^\circ, 26^\circ]$, $z \approx 0$)
LVK GWTC-5 BNS at $z = 0$ (class $\times$ beaming propagated)